

Two hypotheses are tested. (H1) Both CSIL and CSIL+SOAR improve over BC across $K \in \{20, 50, 100\}$ expert trajectories on Acrobot. (H2) SOAR provides no improvement over plain CSIL in the discrete, reduced-budget regime considered here. Hypothesis (H2) is stated a priori: [2] §6 reports SOAR gains only on continuous, higher-dimensional MuJoCo environments at 5M env steps, whereas the setup studied here is small, discrete, and $10\times$ below that budget.

1 Coherent Soft Imitation Learning (CSIL)

CSIL combines behavioural cloning (BC) and inverse reinforcement learning (IRL) by using the BC policy as both a *reward generator* and a *critic initialiser*. The construction relies on the entropy-regularised soft Bellman equation $Q(s, a) = r(s, a) + \gamma \mathbb{E}_{s'}[\mathcal{V}_\alpha(s')]$, whose fixed-point policy admits the Boltzmann form $q_\alpha(a|s) \propto p(a|s) \exp(\alpha^{-1}Q(s, a))$ ([1] Eq. 4) for a prior p and temperature α .

1.1 The coherent reward and its derivation

Inverting the Boltzmann form for Q yields $Q(s, a) = \alpha \log(q_\alpha(a|s)/p(a|s)) + \mathcal{V}_\alpha(s)$ (Theorem 1 of [1]). Applying the Ng–Harada–Russell shaping lemma [4] with potential $\Psi(s) = -\mathcal{V}_\alpha(s)$ cancels the bootstrap, leaving the *coherent reward* (Definition 1 of [1])

$$\bar{r}(s, a) = \alpha \log(q_\alpha(a|s)/p(a|s)), \quad (1)$$

for which q_α is provably soft-optimal at temperature α . For our discrete action spaces we set the prior to uniform, $p(a|s) = 1/|\mathcal{A}|$, so $\log p = -\log |\mathcal{A}|$ is a constant and (1) reduces to a shifted log-likelihood of π_{BC} .

1.2 Three-stage training pipeline (Algorithm 1)

Stage A — BC. $\theta_1 = \arg \max_\theta \mathbb{E}_{\mathcal{D}}[\log q_\theta(a|s)] - \Psi(\theta)$ with weight-decay regulariser Ψ and patience-based ($P=500$) early stop on a held-out cross-entropy split ([1] §L: “if the policy is trained too long, the entropy is reduced too far, and there is not enough exploration for RL”).

Stage B — SARSA warm-up. We initialise Q_ϕ via TD-SARSA on expert transitions with the frozen reward $\bar{r}_{\theta_1}$ and target $y = \bar{r}_{\theta_1}(s, a) + \gamma(1-d)Q_\phi(s', a')$, matching Q_ϕ to π_{BC} before any environment interaction.

Stage C — SAC fine-tune, $\beta < \alpha$. The actor maximises the KL-anchored SAC objective $\mathcal{J}_\pi = \mathbb{E}_{a \sim q_\theta}[Q - \beta(\log q_\theta - \log \pi_{\text{BC}})]$ ([1] Eq. 5); the critic regresses to $y = \bar{r}_{\theta_1} + \gamma(1-d)\mathbb{E}_{a' \sim q_\theta}[Q_\phi(s', a') - \beta(\log q_\theta(a'|s') - \log \pi_{\text{BC}}(a'|s'))]$. The strict inequality $\beta < \alpha$ (asserted at runtime in `CSILAgent.__init__`) is required: at $\beta = \alpha$ the soft-optimal policy of $\bar{r}$ is π_{BC} itself, so Stage C cannot improve on BC. Lowering β loosens the KL anchor and lets SAC exploit environment data.

1.3 Algorithm 2 actor reward refinement

Beyond $\mathcal{J}_\pi$, we use the actor-side reward refinement of [1] Eq. 11–12, which minimises Schulman’s positively-constrained KL estimator:

$$-\mathcal{J}_r(\theta) = -\mathbb{E}_{(s,a) \sim \mathcal{D}}[\bar{r}_\theta] + \mathbb{E}_{(s,a) \sim \rho_\pi}[\bar{r}_\theta - 1 + \exp(-\bar{r}_\theta)]. \quad (2)$$

A separate Adam optimiser ($1r=10^{-4}$) updates θ once per env-iteration. Gradient norms are clipped to 1 and the replay-side reward is clamped to $[-10, +\infty)$ before $\exp(-\bar{r})$, since categorical $\log \pi_{\text{BC}}$ goes to $-\infty$ as $\pi_{\text{BC}}(a|s) \rightarrow 0$ and produced NaN gradients in early Acrobot runs without the clamp.

1.4 Implementation deviations

(i) **Critic reward.** [1] Algorithm 2 uses the *current* θ for $\bar{r}_\theta$ in the critic target. A preliminary single-seed test at $K=50$ showed this closed loop diverging on both environments (CartPole return collapsed $500 \rightarrow 137$; Acrobot $-82 \rightarrow -300$ within 30 Stage-C iterations): the discrete categorical actor cannot absorb the $\exp(-\bar{r})$ feedback at the available training budget. The critic therefore retains the Algorithm 1 frozen- θ_1 reward, and $\mathcal{J}_r$ refines only the actor. (ii) **Reward learning rate** 10^{-4} instead of 10^{-3} : the higher rate destabilises $\mathcal{J}_r$ when categorical log-likelihoods span ~ 10 orders of magnitude on small $|\mathcal{A}|$.

2 IL-SOAR — Optimistic ensemble exploration

IL-SOAR [2] wraps any SAC-based imitation algorithm by replacing the single critic with an L -member ensemble and aggregating into an upper-confidence estimate. Each member ℓ is itself a twin pair

$(Q_\ell^{(1)}, Q_\ell^{(2)})$ trained on the shared backup $y_\ell = \bar{r} + \gamma(1 - d) \min(Q_\ell^{(1)}, Q_\ell^{(2)})$ (Algorithm 7 of [2]; the in-pair min is SAC’s standard anti-overestimation trick).

2.1 Optimistic aggregation

Aggregation uses option 2 (“Mean-Std”) of [2] Algorithm 4. We translate from the cost form ($Q = c + \gamma \max(\bar{Q} - \sigma, 0)$, line 6) to the reward form by flipping the sign analogously to footnote 8’s twin-critic max→min convention, so the actor sees

$$Q_{\text{opt}}(s, a) = \underbrace{\frac{1}{L} \sum_\ell Q_\ell(s, a)}_{\bar{Q}(s, a)} + \text{clip}\left(\sqrt{\frac{1}{L} \sum_\ell (Q_\ell(s, a) - \bar{Q}(s, a))^2}, 0, \sigma_{\text{clip}}\right), \quad (3)$$

with $Q_\ell = \min(Q_\ell^{(1)}, Q_\ell^{(2)})$ and the biased $1/L$ variance estimator (matching [2] Algorithm 5 line 4).

2.2 Bootstrap variant and SOAR-specific deviations

[2] Corollary 4.11 lower-bounds L for high-probability optimism, and the §6.1 ablation (Fig. 4) confirms that $L=1$ fails because the members collapse to identical estimates. To keep the members independent we sample a fresh minibatch per ensemble member at every critic step. In a diagnostic run, a shared minibatch gave $\text{std}_\ell Q \approx 0.07$ and per-member resampling gave ≈ 0.40 , a $5.7\times$ jump; without per-member resampling the UCB bonus is negligible.

Two additional SOAR-specific deviations. (iii) [2] Algorithm 6 (App. F) uses L *independent* replay buffers with L parallel rollouts. A single shared buffer is used here, with per-critic bootstrap sampling, since $L=4$ parallel rollouts exceed the available compute budget. (iv) The SOAR ensemble critics use ReLU and no LayerNorm (default MLP), while the plain-CSIL critic uses ELU with LayerNorm ([1] §L). We did not propagate the architecture to the ensemble; this is a confound for the SOAR-vs-CSIL comparison and we return to it in the Limitations.

Discrete adaptations. The uniform prior makes $\bar{r}$ reduce to $\alpha \log \pi_{\text{BC}} + \text{cst.}$ For minimum-time tasks ($r \equiv -1$, Acrobot), [1] §5.3 recommends subtracting $\sup_a \bar{r}_{\theta_1}(s, a')$ so $\bar{r}$ stays non-positive; the action-independent shift cancels in $\mathcal{J}_\pi$. We enable this on Acrobot only.

3 Experiments

3.1 Setup

Environments. **CartPole-v1** ($|\mathcal{A}| = 2$, return cap 500) and **Acrobot-v1** ($|\mathcal{A}| = 3$, $r \equiv -1$). The expert is PPO from Stable-Baselines3, trained 2×10^5 steps on CartPole and 5×10^6 on Acrobot, with deterministic test-time action selection (stochastic PPO sampling around saturated logits caps BC at ~ -72 on Acrobot).

Conditions. $K \in \{20, 50, 100\}$ expert trajectories, spanning one decade of demonstration budget but staying in the regime where BC plateaus well below the expert (initial sweeps at $K \leq 10$ left BC at ~ -77 , with no headroom for Stage C). Three seeds per cell, both algorithms on Acrobot, 18 runs total. Stage A: 5000 BC steps with early stop. Stage B: 2000 SARSA steps. Stage C: 480 iterations $\times$ 1000 env steps (**480k env steps**, ≈ 6000 episodes), with 16 gradient steps per iteration on each of {actor, critic}. This budget is $\sim 10\times$ below the 5M reported in [1]; the 18 benchmark cells at the full budget would require ~ 7 days of single-GPU compute, beyond the available allocation.

Hyperparameters. $\alpha = 1$, $\gamma = 0.99$, Polyak $\tau = 5 \times 10^{-3}$, actor/critic lr 10^{-3} , reward lr 10^{-4} , batch 256, replay 10^5 , hidden (256, 256). $\beta = 0.05$ (CartPole) / 0.10 (Acrobot): [1] Table 3 reports $\beta = 0.01$ as optimal at 5M steps, but a single-seed sweep at the 100k-step budget showed $\beta = 0.01$ destabilises the critic (the replay- $\mathcal{J}_r$ feedback loop diverges); we adopt the next-larger values in [1] Table 3. $L = 4$: [2] Fig. 4 ablation shows $L = 4$ recovers most of $L = 16$ ’s performance at 1/4 the wall-clock cost. $\sigma_{\text{clip}} = 1.0$ (Acrobot): an upper bound on the UCB bonus in the event of large ensemble disagreement. Empirically $\text{std}_\ell Q < 0.7$ throughout training, so the clip is never active.

Evaluation protocol. Following [1] Fig. 4 caption, the deterministic actor is evaluated on 20 episodes with fixed seeds 10000–10019 at every $n_{\text{iters}}/10 = 48$ Stage-C iterations (and at $t=0$). The primary metric is the 25th percentile Q_{25} of those 20 returns at the *best-snapshot* checkpoint, restored before final evaluation. Q_{25} is preferred over the mean: on $N=20$ episodes a single high-return rollout can shift the mean, whereas Q_{25} reflects the lower quartile and is less sensitive to such outliers. Standard

deviations are reported in their sample form (ddof=1, $N=3$ seeds).

3.2 Main results

We present two complementary views. Fig. 1 plots the per-checkpoint mean return (noisy training dynamics); Fig. 2 plots the per-seed cumulative maximum of Q_{25} then averaged across seeds (the monotone snapshot view of [1] Fig. 4, in which each y-value is the best Q_{25} if training stopped at that episode). Table 1 gives the final snapshot Q_{25} per cell.

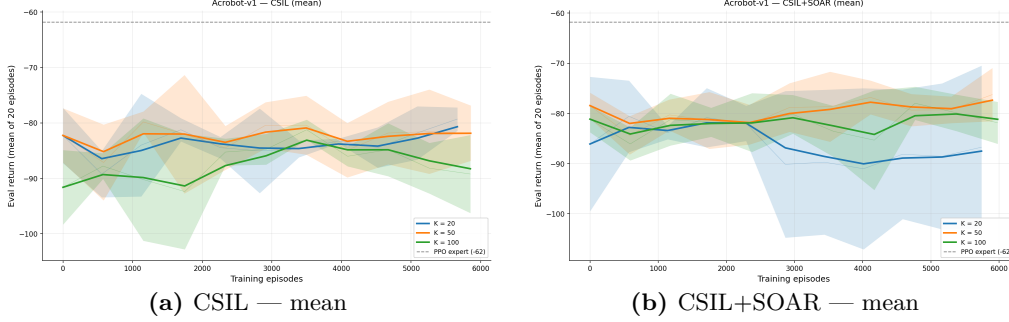


Figure 1: Raw training curves on Acrobot-v1, ≈ 6000 Stage-C episodes. Y: checkpoint mean return over 20 deterministic episodes; X: cumulative training episodes; shaded $\pm$ one ddof=1 std across 3 seeds; dashed: PPO expert (-61.8). **(a)** CSIL’s gain over BC scales with K . **(b)** CSIL+SOAR reaches higher plateaus at $K=50, 100$ but visibly regresses at $K=20$ (blue): one seed’s optimistic bonus pushed the actor away from the BC initialisation, dragging the mean to ~ -90 around episode 4000.

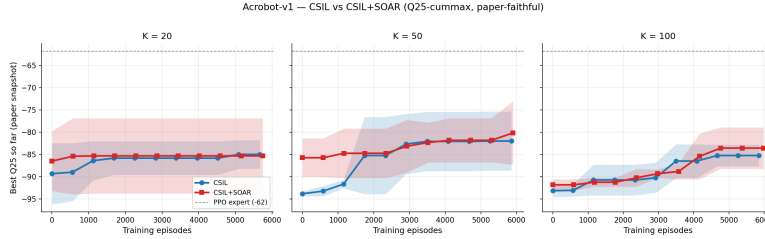


Figure 2: Snapshot view: cumulative maximum of Q_{25} , CSIL vs. CSIL+SOAR. Y: per-seed $\max_{t' \leq t} Q_{25}(t')$, mean across 3 seeds; shaded $\pm$ ddof=1 std; dashed: PPO expert. Curve form of the snapshot protocol ([1] Fig. 4): each y is the best Q_{25} if training stopped at that episode. At $K=50, 100$ CSIL+SOAR (red) plateaus a few hundred episodes earlier than CSIL (blue) but with wider bands; at $K=20$ the two are statistically tied.

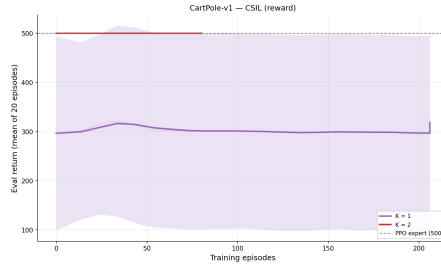


Figure 3: Why CartPole is excluded. CartPole-v1’s state space is low-dimensional and the expert’s policy is near-linear in the observations, so BC absorbs the expert almost perfectly with very little data: at $K=2$ trajectories (~ 400 transitions), BC already saturates the 500/500 return cap (red). With $K=1$ (purple) the policy is still under-data and noisy. Any $K \geq 20$ is therefore trivially capped — Stage C has zero headroom, and the CSIL-vs-SOAR contrast cannot be measured.

Table 1: Q_{25} at the best snapshot following [1] Fig. 4, $\text{mean} \pm \text{ddof}=1$ std over 3 seeds. The *gap* is the improvement over each algorithm’s own BC baseline (BC values differ between CSIL and SOAR runs by ~ 3 pts because the `torch` random number generator advances differently in the two agent constructors; the gap is computed row-wise within the same random seed). All numbers reproduce from `benchmark_csil_Acrobot-v1.csv` and `benchmark_csil_soar_Acrobot-v1.csv` to ± 0.1 . PPO expert: -61.8 ± 0.7 .

	$K=20$		$K=50$		$K=100$	
	Q_{25}	gap	Q_{25}	gap	Q_{25}	gap
BC (CSIL run)	-89.3 ± 6.9	—	-93.8 ± 0.6	—	-93.2 ± 1.5	—
CSIL	-85.0 ± 3.2	+4.3	-82.0 ± 6.6	+11.8	-85.2 ± 2.4	+8.0
BC (SOAR run)	-86.5 ± 6.7	—	-85.8 ± 4.4	—	-91.8 ± 1.2	—
CSIL+SOAR	-85.3 ± 8.5	+1.2	-80.2 ± 7.1	+5.6	-83.6 ± 4.6	+8.2

3.3 Observations

(i) **H1 confirmed — both CSIL and CSIL+SOAR improve BC across all K .** All 6/6 ($\text{algo} \times K$) mean cells in Table 1 are positive; per-seed, 16/18 gaps are strictly positive with 2 ties at $K=20$ and no negatives. The largest gap, +11.8 at $K=50$, falls where you would expect: Stage C has the most room when BC covers enough state-action support to bootstrap from, but not so much that BC already sits at the expert plateau. CSIL snapshot iterations $t^* \in \{0, 96, 144, 240, 288, 384, 432\}$: on 8/9 runs Stage C improves on BC for between 20% and 90% of the budget; on the remaining run (seed 0, $K=20$) Stage C did not beat BC.

(ii) **H2 confirmed — CSIL+SOAR matches CSIL on the mean, with higher variance.** SOAR is marginally better at $K=50$ (+1.8) and $K=100$ (+1.6), tied at $K=20$ (−0.3). Across K , $\bar{\sigma}_{\text{SOAR}}=6.7$ vs. $\bar{\sigma}_{\text{CSIL}}=4.1$. At $K=20$ seed 2, the SOAR snapshot iteration is $t^*=0$: *every* subsequent checkpoint had $Q_{25} < -88.5$, i.e. the optimistic bonus actively prevented improvement in 1/9 runs (visible as the blue curve in Fig. 1b). With $N=3$ seeds this comparison is underpowered; we treat it as suggestive, not conclusive (see §4).

(iii) **~ 20 points remain to the expert.** Best CSIL or SOAR Q_{25} is -80.2 vs. expert -61.8 . Our budget is $\sim 10\times$ below [1]’s 5M; whether the residual gap reflects insufficient interactions or an intrinsic ceiling for our discrete actor is examined in §4.

(iv) **CartPole was excluded** (Fig. 3): on this low-dimensional task BC absorbs the expert’s policy almost immediately ($K=2$ already locks in 500/500 for both algorithms), so any $K \geq 20$ leaves Stage C flat at the return cap with no headroom to discriminate CSIL from CSIL+SOAR.

4 Discussion

Why Stage C curves stay close to BC. The coherent reward $\bar{r} = \alpha \log(\pi_{\text{BC}}/p)$ shared by CSIL and CSIL+SOAR carries *no information about the environmental goal*: its soft-optimal policy at temperature α is exactly π_{BC} (Theorem 1). Stage C can only depart from BC through the $\alpha - \beta$ gap, and $\bar{r}$ gives no direction for that departure. The actor therefore wanders: the raw mean (Fig. 1) takes +5 to +15 point excursions and falls back, while the snapshot view (Fig. 2, [1] Fig. 4 protocol) keeps the peaks. This gap between the two views explains why the Q_{25} values in Table 1 record the +4 to +12 point gains of H1 even when the raw curves look flat. CSIL and CSIL+SOAR both amplify the BC policy; neither searches for a better one.

SOAR yields no improvement over CSIL on Q_{25} , consistent with [2] §6: the ensemble bonus appears only on continuous higher-dimensional MuJoCo environments. Our data do not separate (A) SOAR is intrinsic to continuous high-dim control from (B) SOAR would help here too at 5M env steps; discriminating would require CSIL+SOAR on Acrobot at 5M or on Ant-v5 at 480k, neither done. A second regime factor compounds (B): the discrete softmax actor with $|\mathcal{A}|=3$ and $\beta=0.10$ is near-deterministic ($\sim 98\%$ argmax in diagnostic runs not re-logged), so a UCB bonus of magnitude $\text{std}_\ell Q \approx 0.4$ is small next to the ~ 5 Q -gap between actions.

Limitations. (1) Three seeds per cell; any σ -level claim, including obs (ii), needs ≥ 10 seeds before it can be taken seriously. (2) The $\sigma_{\text{clip}}=1.0$ ceiling on Acrobot never fired in our runs, so we have no

evidence on its true sensitivity; a sweep $\sigma_{\text{clip}} \in \{0.1, 0.3, 0.5, 1.0, 2.0\}$ would settle it. (3) The SOAR ensemble critic uses ReLU and no LayerNorm while plain-CSIL uses ELU and LayerNorm. Every SOAR-vs-CSIL number is confounded by this; we would need to re-run with matched architectures to isolate the SOAR contribution.

References

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